

Company C

Competitive position in the myeloma T-cell-engager and CAR-T field

Anonymized specimen. Comparator, market, regulatory, and IP data are unredacted at primary source. Subject-asset and client identities are redacted per the specimen posture.

Case ref	MYE-2026-04-B	Corpus	MYE-CORPUS-2026Q2-r2313
Subject	Company C (BCMA cell therapy)	Snapshot	2026-04-20 02:11 UTC
Domain	MM TCE / autologous CAR-T	Methodology	MYE-METH-v1.0
Distribution	L2 (Sponsor + IC)	sha256	ac1ede0299b7...

1. At a glance

Company C is an early-stage UK cell-therapy company disclosing a machine-learning platform for CAR binder maturation, positioning around BCMA in multiple myeloma. This file surfaces the primary-source picture across landscape, failure history, comparator efficacy, class safety, IP, financing, publications, and pipeline claims. The reader draws the investment call.

Market	BCMA is the most crowded target in the myeloma cell-therapy landscape: 5 products approved, 210 of 268 (78%) active clinical programs are BCMA-directed.
Failure history	Of 34 programs terminated, withdrawn, or publicly deprioritized 2021–2025, 20 (59%) died of commercial pressure against approved incumbents; 5 (15%) died of construct-level modes the disclosed platform is aimed at.
Efficacy bar	Cilta-cel median PFS 34.9 mo (n=97). Approved TCE class 7.5–11.3 mo. Any new BCMA readout will be measured against these.
Class safety	17,540 unique FAERS cases across the seven approved BCMA and GPRC5D products; 55–69% serious; class is still generating new labeled events (teclistamab IgA nephropathy n=36, IVth-nerve paralysis n=31, both post-approval).
IP adjacency	A co-founder is a named inventor on Celyad's WO2020221873 (BCMA cell therapy, 2019 priority). Max sequence identity from any Company C VHH against Celyad's claimed CDRs is 47%. The six 80%+ hits sit against an unclaimed comparator disclosed but not claimed in the Celyad document.
Financing	One completed priced round (Nov 2022). Three fresh allotment authorities taken since (Jan 2023, Sep 2023, Nov 2025), none filed as completed.
Publication	One NeurIPS workshop preprint on monomer VHH-CAR CDR3 maturation. Reporter-cell activation readout, no primary T-cell functional data in the paper, no multispecific geometry.
Pipeline claim	A multispecific program ("CB-101") with OR-gate logic and Q1 2027 clinic entry appears in exactly one venue (conference speaker bio). No corroboration found across seven venues searched.

What a reader tends to ask next: §2 sizes the crowd; §3 disaggregates the failure surface; §4 places the efficacy bar; §5 shows the class safety profile a new entrant would inherit; §6 walks the IP adjacency at the sequence level; §7 unpacks the published work; §8 reads the Companies House trail; §9 handles the CB-101 single-source claim; §10 lists what would move any of the above over the next 12 months.

Confidence in the underlying evidence: MEDIUM-HIGH. Anchored to primary regulatory, sponsor, patent-office, and Companies House sources. Two items limit lift to HIGH: the FTO adjacency turns on legal claim construction (out of scope for this file); the CB-101 program is unverified in any second venue.

2. Landscape

The myeloma T-cell redirecting market is populated. **Five products are approved for BCMA** (Carvykti, Abecma, Tecvayli, Elrexio, Lynozyfic) and one for GPRC5D (Talvey). The pipeline underneath is skewed: **210 of the 268 active clinical programs (78%) are BCMA-directed**. GPRC5D has 47 active, CD38 has 30, FcRH5 is a single-sponsor target (Roche's cevostamab, five trials). Company C's disclosed asset targets BCMA. It would enter as the 211th BCMA program and, if it reached market, the sixth approved-competitor challenger, all before a first-in-human trial.

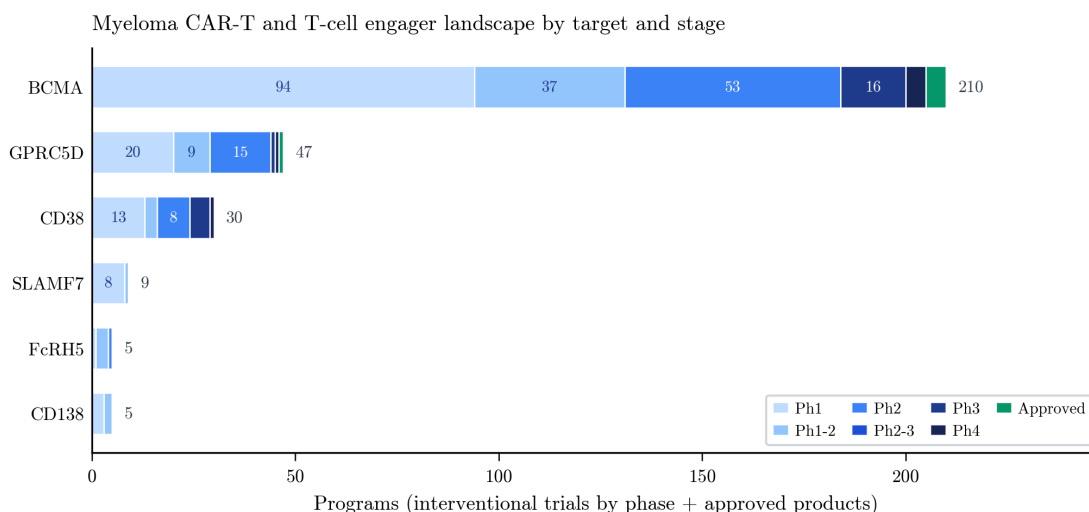


Figure 1: Active interventional myeloma CAR-T and T-cell engager programs by target and phase, plus approved products. *Approved (Drugs@FDA, as of 2026-04-20): BCMA: Carvykti (cilta-cel), Abecma (ide-cel), Tecvayli (teclistamab), Elrexio (elranatamab), Lynozyfic (linvoseltamab); GPRC5D: Talvey (talquetamab).*

Any differentiation thesis for a new entrant in this market operates on construct-level performance (potency, durability, safety, or manufacturing cost per dose) against products that already work. §3 disaggregates the recent failure surface to show which of those levers the field's actual deaths correspond to.

3. Where the field actually fails

Thirty-four myeloma T-cell engager and CAR-T programs were terminated, withdrawn, held, or publicly deprioritized between 2021 and 2025. The distribution of causes:

Twenty of the 34 (59%) exited under commercial pressure. Programs with working efficacy data were shelved on inability to differentiate against approved incumbents. Amgen's AMG 701 (83% ORR at top dose) exited with the whole Amgen BCMA BiTE franchise. Novartis's PHE885 enrolled 146 patients and reported 100% ORR in the

top-dose cohort, then terminated. Legend's BCMA follow-on cited "strategic realignment". Six such exits appear in the corpus.

Four (12%) halted on enrollment once standard of care improved. BMS/2seventy's KarMMa-9 halted at ~10% enrollment across 18 countries over more than a year: patients would not randomize to first-line CAR-T once induction became good enough. This is the observable ceiling on how far into the treatment sequence a BCMA cell therapy has been able to push under current SOC.

Five (15%) fell in the construct-level class a platform of Company C's disclosed kind acts on. Insufficient ORR (four events) and cell persistence failure (one event, Kite/Gilead's KITE-585, which wrote off \$820M for minimal in-vivo expansion). The remaining nine were autologous manufacturing failure (three), unattributed serious tox (two), and format-specific issues (four).

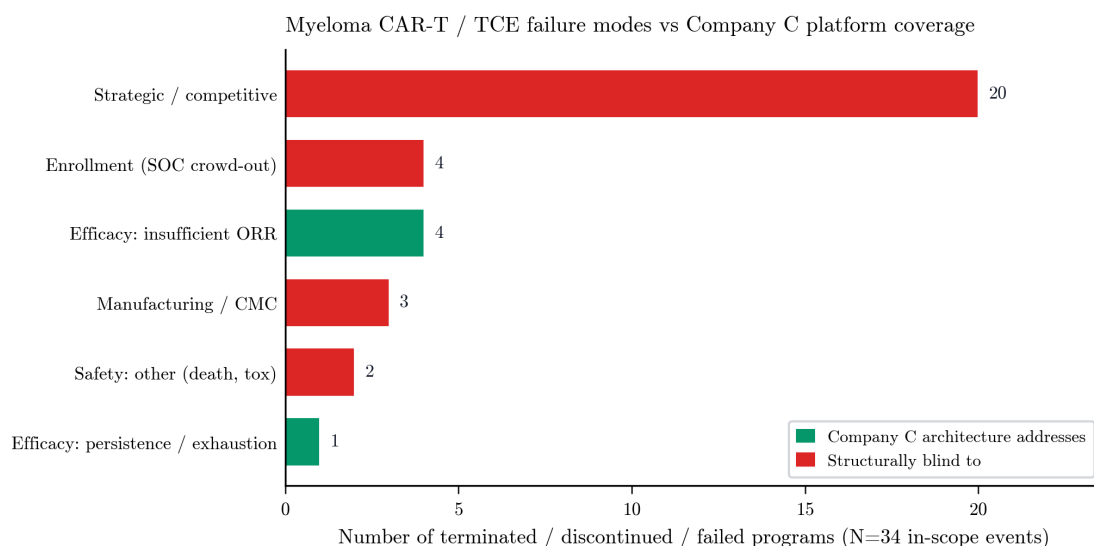


Figure 2: Primary failure mode for each of the 34 events. Green = the mode Company C's disclosed platform can address (construct-level: binder quality and cell persistence). Red = modes the platform does not act on (commercial pressure, patient recruitment, manufacturing, unattributed tox, format-specific issues).

Company C's disclosed platform (in-cell screening + ML-guided sequence optimization for binder maturation) intersects the 15% construct-level slice of this surface. The 85% balance (commercial pressure, SOC crowd-out, manufacturing, unattributed tox, format-specific) is outside what a binder-maturation platform acts on.

4. The competitive benchmark: progression-free survival

Any Company C efficacy read will be compared against the approved BCMA and GPRC5D products at the same follow-up windows. Those benchmarks are set:

Product / setting	n	Median PFS	RMST 12 mo	RMST 24 mo
Cilta-cel (Carvykti), CARTITUDE-1	97	34.9 mo	10.5 mo	18.6 mo
Ide-cel (Abecma), KarMMa-3	254	13.8 mo	8.9 mo	13.8 mo
Standard of care, KarMMa-3 control	132	4.4 mo	6.0 mo	8.2 mo
Teclistamab (Tecvayli), MajesTEC-1	165	11.3 mo	7.9 mo	n/a
Talquetamab (Talvey) 0.8 mg/kg Q2W, MonumenTAL-1	154	11.2 mo	8.0 mo	12.4 mo
Talquetamab 0.4 mg/kg QW, MonumenTAL-1	143	7.5 mo	7.4 mo	10.8 mo
Talquetamab prior TCR, MonumenTAL-1	78	7.7 mo	7.3 mo	11.6 mo

Two features of this table. First, cilta-cel's median PFS is roughly 3× the approved TCE class. A head-to-head efficacy claim from a new BCMA entrant is read against cilta-cel; a claim in the 11–14 month range sits in Abecma / Tecvayli / Talvey territory. Second, teclistamab's 24-month risk set is empty in MajesTEC-1, so no mature 24-month benchmark exists for the TCE class. A durability read at that window from a new entrant would have no direct comparator on the approved TCE side; against Carvykti's 34.9-month median the comparator is already there.

Restricted mean survival time (RMST) reads the same distribution differently: the 24-month RMST spread across approved products is 8 to 19 months.

5. Class safety in the wild

Six years into the approved class, the post-market picture is stable and informative for a new entrant. FAERS reporting counts for the seven approved BCMA and GPRC5D products through 2026Q2:

Product	Cases	2026Q2	% serious	Deaths	Emergent PTs
Cilta-cel (Carvykti)	5,877	791	63.9%	749	495
Teclistamab (Tecvayli)	3,619	243	55.8%	755	288
Belantamab mafodotin (Blenrep)	2,934	174	62.4%	1,000	83
Elranatamab (Elrexio)	2,162	298	69.4%	510	426
Talquetamab (Talvey)	2,056	210	49.2%	228	362
Ide-cel (Abecma)	1,225	63	51.9%	173	133
Linoseltamab (Lynozyfic)	56	20	59.4%	3	75

Raw cross-product counts track exposure volume, not comparative risk. The class-level pattern of interest is emergent-term generation: teclistamab's post-2025Q1 emergent terms include **IgA nephropathy** (n=36) and **IVth nerve paralysis** (n=31), neither on the label at approval; cilta-cel's include lymphoproliferative disorder (n=15). Low-frequency, long-latency events of this shape typically appear in label updates over the following 18–24 months and set the class-level expectations any new BCMA entrant's own package is read against.

Two structural notes on the counts. Belantamab fell from a 2022 peak of 364 to 8 in 2025Q2 and rose back to 174 by 2026Q2, consistent with the 2022 US market withdrawal and partial rehabilitation. Linvoseltamab is too new to compare (56 cases total). Ide-cel's absolute case count is falling as cilta-cel takes share, with a constant death-outcome fraction.

6. Intellectual property and freedom to operate

A Company C co-founder is a named inventor on **WO2020221873**, an anti-BCMA cell therapy patent assigned to Celyad Oncology with a 2019 priority. The founder overlap surfaces the question of whether any Company C sequence reads on any Celyad claim. The document has two kinds of sequence material inside it.

Celyad's claimed antibody in WO2020221873 is a conventional VH/VL scFv. Company C uses VHH format. The only VHH-shaped material anywhere in Celyad's document is a comparator construct that Celyad discloses but does not claim. The relevant comparison is Company C VHH sequences against Celyad's claims, not against Celyad's disclosed but unclaimed comparator.

Against the CDRs Celyad actually claims, the maximum identity from any Company C sequence is **47%**. The six 80%+ hits are all in the CDR2 position and all against the unclaimed Celyad comparator:

Company C CDR	Celyad match	Sequences	% id	Basis
SEQ ID NO: 172	SEQ ID NO: 15 (unclaimed)	IAGSGGST vs IAWSGGST	87.5%	IMGT CDR2, 8 aa
SEQ ID NO: 172	SEQ ID NO: 15 (unclaimed)	17-aa Kabat window	82.4%	Kabat CDR2, framework-inflated
SEQ ID NO: 67 / 68	SEQ ID NO: 15 (unclaimed)	ITSGGST vs IAWSGGST	85.7%	IMGT CDR2, shorter-seq basis
SEQ ID NO: 67 / 68	SEQ ID NO: 15 (unclaimed)	13-of-17 Kabat window	81.3%	Kabat CDR2, shorter-seq basis

Two structural notes on the numbers. The top hit's Kabat window pulls in VHH framework-2 residues (YYADSVKG) that are essentially invariant across the VHH scaffold, so a portion of the identity reflects shared architecture rather than paratope similarity. The CDR2 motif itself (xGSGGST) recurs across multiple unrelated Company C binders in the same sequence listing.

The IP question that remains open is legal, not sequence-level: whether Celyad's claim scope reaches the unclaimed comparator, whether the shared CDR2 motif reads as a paratope under claim construction, and whether framework-2 identity is legally relevant. An FTO opinion at seed / Series A is standard practice regardless; those are the three questions such an opinion would address here.

7. What the published work actually shows

Company C has one technical disclosure in the public record: a workshop preprint at NeurIPS 2024 on machine-learning-guided hit maturation of monomer VHH-CAR binders. Three observations from a direct read of the paper.

Reported metrics. Model loss correlates with reporter-cell activation at Pearson $r = -0.57$ to -0.70 ; top ML-

selected mutants roughly double reporter activation over the parental construct. Both are stated in the paper as measured.

Scope of the paper. Monomer VHH-CAR CDR3 maturation. The activation readout is a reporter cell, not a primary human T cell. The paper does not describe multispecific antibody architectures, linker optimization, valency selection, or the ML loop applied to anything beyond CDR3. Framings appearing outside the paper (in third-party summaries and in Company C’s own conference-stage pipeline description) that claim any of those are outside the published disclosure.

Publication status. Workshop preprint at NeurIPS. Not journal peer review. Consistent with an early-stage company’s disclosure posture, and worth naming precisely.

§10 lists the specific disclosures that would extend the published record beyond this window.

8. Financing state and cap-table signals

Company C completed one priced equity round, in November 2022. Both founders’ Persons-with-Significant-Control status ended on the same date as the share allotment, and an investor-associated director joined the board. Round size and valuation are not filed at Companies House (the register shows nominal share capital only). Since November 2022 the company has taken three fresh allotment authorities from shareholders (January 2023, September 2023, and November 2025) and *completed none of them*. Standing authority to issue shares is not equivalent to shares issued. The most recent authority (November 2025) is roughly five months before the corpus snapshot; the pattern from the register is consistent with either a company preparing rounds that did not close, or one raising via convertible instruments that have not yet converted, or one simply renewing unused authority. It is not consistent with a company that has closed a priced round since 2022.

Event	Filing	Date	Outcome
Priced round	Allotment + PSC transitions + director appointment	Nov 2022	completed
Fresh allotment authority	RES10 + RES11	Jan 2023	no allotment filed
Fresh allotment authority	RES10 + RES11	Sep 2023	no allotment filed
Board change	Investor-associated director appointed	Apr 2025	completed
Fresh allotment authority	RES10 + RES11	Nov 2025	no allotment filed (as of snapshot)

The register currently carries an active statement that the company believes there is **no registrable person** (no shareholder with more than 25% control). This is the state that would follow the November 2022 round pushing the largest holder below the threshold with no subsequent event concentrating control back above it.

9. Single-source pipeline claim

A multispecific T-cell engager program (“CB-101”) with OR-gate logic and Q1 2027 clinic entry appears in exactly one venue: a conference speaker-bio attribution on a page now behind a login redirect. Seven independent venues were searched for corroboration (sponsor website and Wayback captures, SEC EDGAR, MHRA and FDA filings, WIPO PATENTSCOPE composition-of-matter searches, preprint servers, seven major immuno-oncology and biotech conference agendas, sponsor social posts); no corroboration surfaced in any of them. This file treats the claim as unverified and no other section relies on it. A second-venue appearance would upgrade it.

10. 12-month watchlist

1. **Preclinical data with a primary T-cell functional readout.** *If it appears:* platform generalization beyond reporter Δ GFP is demonstrated. *If it does not:* the reporter-vs-primary gap in §7 remains open.
2. **ML loop extended to multispecific architecture.** *Appears:* architectural scope reaches the LabGenius comparator; CB-101 framing has published support. *Does not:* the multispecific framing stays positioning-level.
3. **CB-101 corroboration in any second venue** (press release, conference abstract, IND filing, patent). *Appears:* the pipeline claim moves from single-source to verifiable.
4. **Next SH01 filing at Companies House.** *Appears:* the standing allotment authority converted; runway extended. *Does not:* authority taken but unused.
5. **FDA or MHRA IND cover-sheet appearance.** *Appears:* clinical transition begins.
6. **Out-licensing, partnership, or services deal disclosed.** *Appears:* a second revenue path is validated; single-asset concentration reduces.
7. **Additional BCMA-program deprioritizations in the 210-program pipeline.** *Appears:* the saturation pattern from §3 continues.

Item 1 is the highest-information disclosure on the list. It is monitorable at venues the subject asset already uses (NeurIPS, AACR, SITC, arXiv), and it bears directly on the reporter-versus-primary observation in §7.